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**Semester:** III

**Session:** 2024-2028

**Course Title:** Computer Organization and  
Assembly Language

**Date of Submission:** 30<sup>th</sup> SEP 2025

**QUESTION 1:** AX contains a non-zero number. Count the number of ones in it and store the result back in AX. Repeat the process on the result (AX) until AX contains one. Calculate in BX the number of iterations it took to make AX one. For example BX should contain 2 in the following case:

AX = 1100 0101 1010 0011 (input – 8 ones)

AX = 0000 0000 0000 1000 (after first iteration – 1 one)

AX = 0000 0000 0000 0001 (after second iteration – 1 one)  
STOP

## **SOLUTION:**

[org 0x0100]

start:

mov ax, 0C5A3h

xor bx, bx

repeat\_process:

```
xor dx, dx      ; dx is 0
mov cx, 16      ; cx is 16
```

count\_bits:

```
shr ax, 1      ; shift AX right
jnc skip_inc   ; if CF = 0, skip increment
inc dx         ; dx = dx +1
```

skip\_inc:

```
loop count_bits ; loops 16 times
```

```
mov ax, dx     ; store number of 1s in ax
inc bx         ; increment iteration counter
cmp ax, 1
jne repeat_process ; if AX != 1, repeat the process
```

```
; AX = 1, BX = number of iterations
```

```
mov ah, 4Ch
```

```
int 21h
```

## QUESTION 2:

Write a program to add two 64 bit numbers given below.

Num1 = 1001 0101 0011 1001 0100 1000 1000 0000 1001 0101  
0011 1001

0100 1000 1000 0000

Num2 = 1011 1001 0111 1011 0101 1010 0000 1000 1011 1001  
0111 1011

0101 1010 0000 1000

## SOLUTION:

```
[org 0x0100]
```

```
jmp start
```

```
num1:  dw 9539h, 4880h, 9539h, 4880h
```

```
num2:  dw 0B97Bh, 05A08h, 0B97Bh, 05A08h
```

```
result: dw 0,0,0,0
```

```
start:
```

```
    mov si, num1
```

```
    mov di, num2
```

```
mov bx, result
```

```
mov cx, 4      ; 4 words = 128 bits
```

```
next_word:
```

```
mov ax, [si]
```

```
adc ax, [di]
```

```
mov [bx], ax
```

```
add si, 2      ; increments 2
```

```
add di, 2
```

```
add bx, 2
```

```
loop next_word
```

```
mov ah, 4ch
```

```
int 21h
```